Hosting Capacity and the Data-Center Connection Game

How much load can a bus take, and who gets it?

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Why this question, now

- Large new loads, especially AI data centers, ask for hundreds of MW at one point of the transmission grid.
- Before a request is approved, the utility must answer: how much load can this bus take without violating network limits?
- When several requests compete for the same headroom: who gets served, and how much?

Today

1. Hosting capacity as an optimization problem: one bus, many periods, many buses, with local generation and storage, with flexibility.
2. The connection game: a classroom version of this planning problem on a 2,000-bus synthetic Texas grid.

Hosting capacity

What is hosting capacity?

Definition

The largest amount of new generation or *load* that can be connected without performance becoming unacceptable, with the existing network and controls. [Bollen & Hassan 2011; Bollen & Rönnberg 2017]

- EPRI: "the amount of PV that can be accommodated without impacting power quality or reliability under existing control and infrastructure configurations" [Rylander et al., EPRI 2015]
- Originally for PV on distribution feeders; the same idea applies to large loads on transmission [Cicilio et al. 2021; Wan et al. 2026].
- Ways to compute it: stochastic (Monte Carlo), streamlined, time-series, and optimization-based [Mulenga et al. 2020; Ismael et al. 2019].

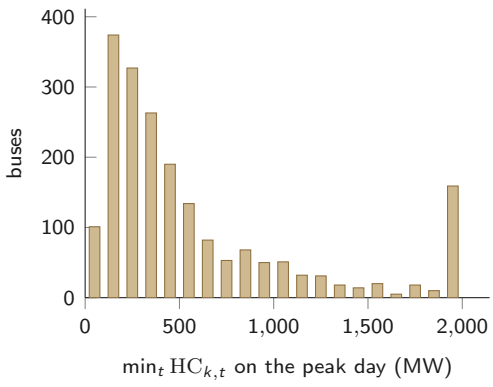
Single-bus load hosting capacity as an LP

Extra load L at bus k in period t; existing generators redispatch:

$$\begin{aligned}
\mathrm{HC}_{k,t} = \max_{L,\,p,\,\theta} \quad & L \\
\text{s.t.} \quad & \textstyle\sum_{g\in\mathcal{G}_n} p_g + s_{n,t} - d_{n,t} - L\,\mathbf{1}[n=k] = \sum_{\ell:\,o(\ell)=n} f_\ell - \sum_{\ell:\,r(\ell)=n} f_\ell \quad \forall n \\
& f_\ell = b_\ell(\theta_{o(\ell)} - \theta_{r(\ell)}), \quad |f_\ell| \le F_\ell \quad \forall \ell \\
& p_g^{\min} \le p_g \le p_g^{\max} \quad \forall g, \qquad \theta_{\mathrm{ref}} = 0, \quad L \ge 0
\end{aligned}$$

- The DCOPF constraints with one extra load; the cost objective is replaced by "make L as large as possible" [Alturki et al. 2018; Wan et al. 2026].
- At the optimum a line or the generation limit binds: the bottleneck of bus k (read off the duals).
- PTDF view: $|f_\ell(L)| = |f_\ell^0 - \mathrm{PTDF}_{\ell,k} L + \dots| \le F_\ell$, so buses "far" from tight lines host more.

Hosting capacity varies a lot across buses

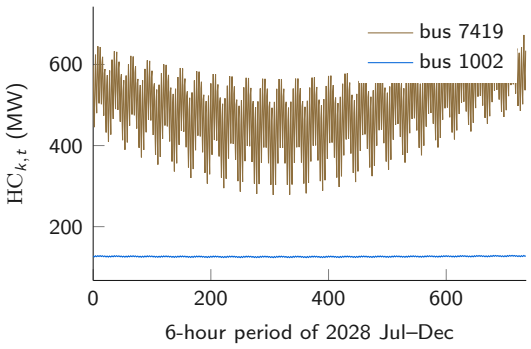


ACTIVSg2000, 2028 Jul–Dec, peak-load day (four 6-hour periods), all 2,000 buses:

- median 379 MW
- 10% of buses below 134 MW
- 90% of buses below 1,543 MW; 148 buses above 2,000 MW (not shown)
- smallest: 10 MW

Computed with the course tool `ercot-game-hosting-capacity` (Gurobi, 8,000 LPs).

Over time: the stressed periods bind



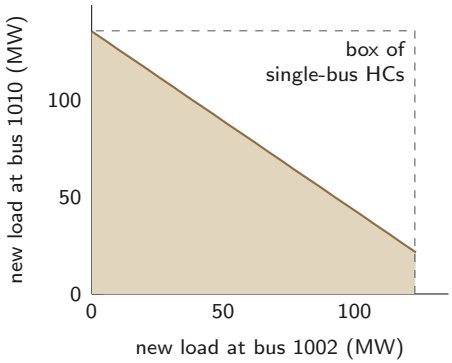
A constant load must be served in *every* period:

$$\mathrm{HC}_k = \min_t \mathrm{HC}_{k,t}$$

- Bus 7419: min 278, median 502 MW (load-driven: tight in peak periods)
- Bus 1002: min 124, median 126 MW (structural: almost constant)

"Dynamic hosting capacity" treats this time variation explicitly [de Oliveira et al. 2019].

Many buses at once: the hosting-capacity region



$$\mathcal{H} = \{x \geq 0 : \exists\, p_t \text{ feasible with loads } d_t + x \;\; \forall t\}$$

- A convex polytope (projection of the DCOPF feasible set) [Gong & Cobben 2024].
- Single-bus HCs are its axis intercepts; they do not add up [Wan et al. 2026].
- Peak period: bus 1002 hosts 123 MW alone, but only 82 MW once 60 MW sit at neighbor 1010.

With a local generator

Site at bus k: load L, private generator $0 \le g_t \le G$. The grid only sees the *net* withdrawal $L - g_t$:

$$\text{feasible} \iff \exists\, g_t \in [0, G] :\ L - g_t \le \mathrm{HC}_{k,t}\ \ \forall t \quad \Longrightarrow \quad \mathrm{HC}_k^{\mathcal{G}} = \min_t \mathrm{HC}_{k,t} + G.$$

- Exact when the generator is always available; with outages, ramping, or fuel limits, $\mathrm{HC}_k^{\mathcal{G}} \le \mathrm{HC}_k + G$.
- It runs only when the grid is short or when its cost (\$80–250/MWh in the game) is below the local price.
- Buying headroom: in the game \$1.5M per MW of generator, versus \$25M per MW of site that would otherwise go unserved.

With storage: periods become coupled

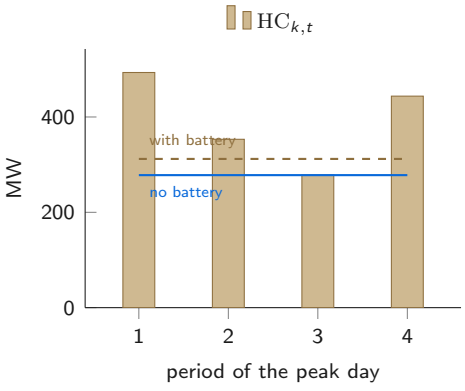
Battery at bus k: power P, energy B, signed output p_t (positive = discharge), period length Δt:

$$\mathrm{HC}_k^S = \max_{L,\,p,\,e} L \quad \text{s.t.} \quad L - p_t \le \mathrm{HC}_{k,t}, \qquad -P \le p_t \le P,$$
$$e_t = e_{t-1} - \Delta t\, p_t, \qquad 0 \le e_t \le B, \qquad e_0 = e_T.$$

- A multi-period LP: the battery discharges in tight periods and must recharge in slack ones, using the same headroom.
- Summing over a cycle ($\sum_t p_t = 0$) gives $\mathrm{HC}_k^S \le \frac{1}{T}\sum_t \mathrm{HC}_{k,t}$: storage can at best lift the limit from the *minimum* to the *average* headroom.
- Power ($L \le \mathrm{HC}_{k,t} + P$) and energy (the running deficit must fit in the battery) keep it below that.

[Macedo et al. 2017 (storage to raise HC); general formulation]

Storage in the game: one day, four 6-hour periods

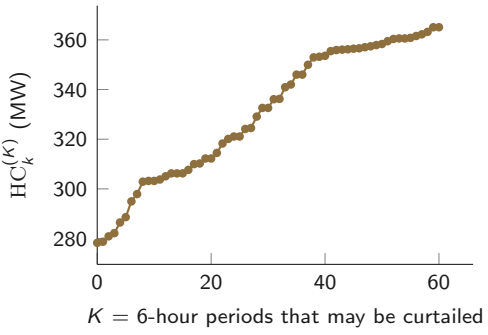


Game rules: $\Delta t = 6\,\mathrm{h}$, $B = 4P$, each day starts and ends at $e = B/2$. Bus 7419, peak day:

- no battery: $278\,\mathrm{MW}$
- average headroom: $392\,\mathrm{MW}$ (upper limit)
- with $P = 50\,\mathrm{MW}$, $B = 200\,\mathrm{MWh}$: $312\,\mathrm{MW}$

A 6-hour period at full power would need $6P > B = 4P$ MWh, so the energy limit binds: even fully charged beforehand, the battery adds at most $B/\Delta t = \frac{2}{3}P$ in one tight period (here $+33\,\mathrm{MW}$). Over the half-year: $278 \to 312\,\mathrm{MW}$.

Flexibility: curtail in a few periods



Flexible connection [Gu, Chen & Qin 2026]:

$$\max_{C,u} C$$
$$\text{s.t. } C + u_t \le \mathrm{HC}_{k,t}, \quad u \le 0, \quad \|u\|_0 \le K$$

Nonconvex ($\|u\|_0$), but for one bus:

$$\mathrm{HC}_k^{(K)} = \mathrm{HC}_{k,[K+1]},$$

the $(K+1)$-th *smallest* value. Bus 7419: 278 MW at $K = 0$, 303 MW at $K = 10$ (60 h, +9%), 354 MW at $K = 40$ (+27%). Diminishing returns in K; on a distribution feeder Gu et al. report +78% at 1% of slots. See also [Etherden & Bollen 2011; Sarker & Dahlin 2026].

How much should we trust a hosting-capacity number?

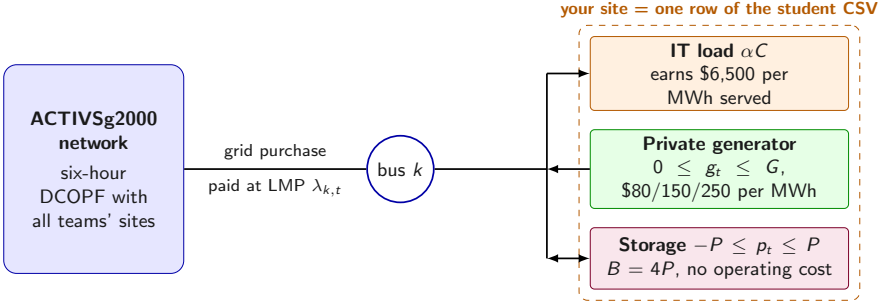
- **Security.** Dropping N-1 contingencies overstates data-center HC by about 20% on the IEEE RTS-24 [Wan et al. 2026].
- **Dynamics.** The data center's own ride-through limits (UPS, converters) can be the binding constraint [Ren, Sun & Teng 2026].
- **Model.** DC power flow has no voltage or reactive-power limits and no losses.
- **Time resolution.** Averaging hours into longer periods smooths peaks and can overstate HC.
- **Simultaneity.** Single-bus numbers ignore everyone else who is connecting at the same time.

The last point is what the game is about.

The connection game

The game

- Teams are data-center operators on ACTIVSg2000 (2,000 buses, 3,206 branches).
- One CSV row per site: bus, connection C, generator $G \leq C$, battery P with $B = 4P$.
- Six-hour periods (mean of 6 source hours), 4 per day, 724–736 per half-year.
- The instructor solves one joint DCOPF with every team's sites.



Schedule and scoring

Week	Submissions	Counts?
1	2027 Jan–Jun, Jul–Dec	practice
2	2028 Jan–Jun, Jul–Dec	both
3	2029 Jan–Jun, Jul–Dec	both

- Sites built in Submission 1 stay for Submission 2 and are not charged again.
- Submission 1 pays no energy cost.

$$\text{score} = \text{IT value} - \text{energy} - \text{gen. operating} - \text{capital}$$

IT value	\$6,500 per MWh served
energy	LMP $\times$ grid purchase
generator operation	\$80 / 150 / 250 per MWh
site	\$25M per MW
generator capital	\$1.5M per MW
storage capital	\$0.4M/MW + \$0.25M/MWh

Break-even share of load served

about 44% over a week if built in Submission 1; 87% if added in Submission 2.

The evaluator

For each day d and period $t = 1, \dots, 4$ ($\Delta t = 6\,\text{h}$): a DCOPF with all teams' sites, coupled through each battery's daily cycle.

$$\min \ \sum_{d,t} \Delta t \Big(\sum_{g} c_g p_{g,dt} + \sum_{s} \sum_{m} c^m q^m_{s,dt} \Big)$$

s.t. DCOPF balance and line limits with loads $\alpha_i C_s$ and private q_s, u_s

$$e_{s,d,t} = e_{s,d,t-1} - \Delta t \, u_{s,dt}, \quad 0 \le e \le B_s, \quad |u| \le P_s, \quad e_{s,d,0} = e_{s,d,4} = B_s/2$$

$$\sum_{s \in i,\, k(s)=n} u_{s,dt} \le \sum_{s \in i,\, k(s)=n} \alpha_i C_s \quad \text{(a team's assets serve only its own load)}$$

- An exact LP that separates by day (Benders or one LP per day).
- LMP = dual of nodal balance; it prices energy in Submission 2.

When requests exceed the hosting-capacity region

The evaluator never uses first-come-first-served. It projects onto the region:

1. **Caps:** $100\,\mathrm{MW}$ per site, $500\,\mathrm{MW}$ per team, largest team capped at the smallest positive team request $\Rightarrow \hat{d}_i$.
2. **Most service:** choose $\alpha_i \in [10^{-6}, 1]$ maximizing $\sum_i \alpha_i \hat{d}_i$ subject to joint feasibility in every period.
3. **Proportional fairness** among near-optimal solutions [Kelly, Maulloo & Tan 1998]:

$$\max \sum_i \log(\alpha_i \hat{d}_i).$$

- Capital is charged on what you *asked* for, not on what is served.
- Alternatives worth discussing: first-come-first-served (today's interconnection queues), max–min fairness, prices [Liu et al. 2020; Gebbran et al. 2021].

A two-team example on the real grid

Team A at bus 1002, team B at neighbor 1010; each picks 50 or 100 MW, built in Submission 1. Peak period, every period treated alike; payoffs per week ($M):

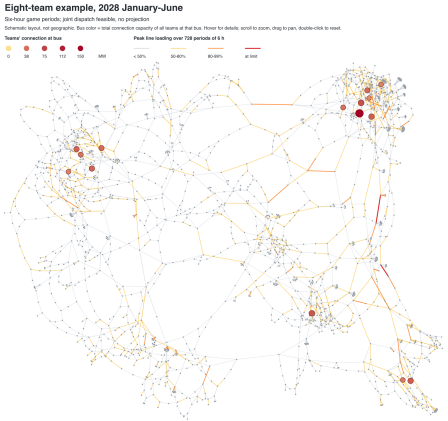
A \ B	B small (50)	B large (100)
A small (50)	1,601 / 1,601	1,601 / 351
A large (100)	351 / 1,601	3,202 / -39

- One large, one small: infeasible; the caps cut the large request to 50 MW, but it still pays for 100 MW.
- Both large: projection maximizes total service; the bus that uses less of the shared bottleneck is favored (A 100 MW, B 43.2 MW).
- Nash equilibrium: (A small, B small).

What students get, and a live demo

- `ercot-game-check-csv`: format and bus checks
- `ercot-game-preview`: approximate score, plan alone
- `ercot-game-hosting-capacity`: the LP from this lecture
- After each round: score breakdown, every team's α, projection on/off, and a grid map

```
ercot-game-hosting-capacity \
  --artifact-dir .../2028/jul_dec \
  --start-period 304 --period-count 4 \
  --bus-id 1002 --bus-id 1010 \
  --output-dir demo/hc
```



Eight-team example: bus color = total MW of all teams; line color = peak loading.

Takeaways

- Hosting capacity is an LP: the DCOPF with a "maximize the new load" objective.
- Time: a constant load is limited by the worst period. Generation adds G; storage moves the limit toward the average; flexibility toward the $(K+1)$-th worst.
- Space: single-bus numbers are intercepts of a region and do not add up.
- With many applicants, allocation is a design choice (queue, proportional fairness, prices), and it turns planning into a game.

Discussion. How should a utility allocate scarce headroom among data centers? What would you change in the game's projection rule?

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